

Priyansh Pathak

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EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science & Engineering - AI/ML Specialization | CGPA: 9.23 / 10

Chennai, India

Aug 2023 - May 2027

Indian Institute of Technology Ropar

Major in Artificial Intelligence

Remote

Sept 2024 - Feb 2026

RESEARCH & INDUSTRY EXPERIENCE

Carnegie Mellon University - Xu Lab

Research Intern - Robot Learning & Embodied AI

Remote

April 2026 - Sept 2026

- Developing educational Jupyter notebooks on Diffusion Policy and RLBench for the xulabs/edu repository, making state-of-the-art robot learning methods accessible to new researchers.
- Implementing and documenting hands-on experiments covering imitation learning, task and motion planning, and manipulation benchmarks using RLBench simulation environments.
- Contributing to open curriculum infrastructure via GitHub PR workflows, with focus on reproducibility, clarity, and pedagogical structure for graduate-level robotics courses.
- Exploring integration of vision-language models with robot manipulation pipelines for generalizable task execution.

Indian Institute of Technology Ropar

Research Intern - School of AI & Data Engineering (Advisor: Dr. Arun Kumar Sharma)

Ropar, India

Jun 2026 - Aug 2026

- Researched bacteriophage-host interaction prediction using deep learning on protein sequence datasets, at the intersection of AI, deep learning, and computational biology.
- Designed and conducted experiments on protein sequence data; implemented the research methodology end-to-end and generated and evaluated experimental results.
- Compared performance across approaches, analyzed outcomes, and derived observations to assess model effectiveness on the phage-host interaction task.

QX Global Group

AI/ML Intern - Invoice Document Automation

Remote

May 2026 - Jul 2026

- Built an invoice verification pipeline converting raw scanned documents into structured, usable data: OCR labelling, text extraction, and downstream OCR model training on the labelled data.
- Used LayoutLM to incorporate layout-aware structure into document understanding, handling real-world scanned documents beyond plain text extraction.
- Worked through messy, real-world document variation not represented in clean benchmark datasets, reinforcing that labelling quality is the primary driver of downstream model quality.

DRDO CAIR

Research Intern - AI Navigation & Path Planning

Remote

Jan 2026 - March 2026

- Developing spatially-aware ML models for intelligent route planning in defense navigation systems.
- Implementing and benchmarking graph-based and reinforcement learning algorithms for path optimization under real-world environmental constraints.
- Reproducing and extending algorithms from recent research papers to establish comparative baselines.

Indian Institute of Technology Mandi

Research Intern - Computer Vision & Deep Learning

Mandi, HP

May 2025 - Jul 2025

- Built an end-to-end video-based attendance system using deep learning: MTCNN for face detection, Siamese networks with triplet loss for robust facial embedding learning.
- Performed comparative evaluation against baseline face recognition models; conducted dataset preprocessing and alignment for classroom video streams.
- Investigated improvements in metric learning for low-quality, unconstrained video conditions.

AIEnsured

AI Research Intern - Fairness & Robustness

Remote

Jul 2025 - Sept 2025

- Built model evaluation pipelines focused on fairness metrics, bias detection, and explainability across multiple datasets.
- Conducted experimental analysis of model behaviour under distribution shift and label imbalance conditions.

Skybrisk Technologies

Remote

Machine Learning Intern

Feb 2025 - Aug 2025

- Developed predictive models and end-to-end data preprocessing pipelines for real-world structured datasets.
- Built AI-driven decision systems, contributing to production-ready deployments.

CodTech IT Solutions

Remote

Computer Vision Intern

Dec 2024 - Jan 2025

- Implemented CNN-based image classification pipelines using TensorFlow/Keras with augmentation and evaluation.

PUBLICATIONS

AttendEase: Video-Based Classroom Attendance using Siamese Networks

AMISE 5.0

Best Paper Award, Accepted; In Publishing Process

Deep Learning-Based Skin Cancer Detection with Uncertainty Quantification

MICAD 2026

Accepted; MICCAI-Endorsed Event, Springer LNEE

PROJECTS

Video-Based Face Recognition Attendance System | PyTorch, MTCNN, Triplet Loss, OpenCV

- Designed a production-grade attendance pipeline ingesting classroom video streams; MTCNN for face detection and Siamese networks with triplet loss for embedding-based recognition.
- Automated attendance logging with CSV storage; evaluated against standard face recognition baselines under low-resolution, real-world conditions.

Skin Cancer Detection Web Application | TensorFlow, EfficientNetB3, Flask, Grad-CAM

- Built a dermoscopic image classifier using EfficientNetB3 with transfer learning; achieved AUC 0.826, sensitivity 0.916 across multi-class lesion categories.
- Integrated Grad-CAM visual explanations and deployed as a Flask web application with preprocessing pipeline fixes for rescaling and Keras 3 compatibility.

Real-Time Face Mask Detection | TensorFlow, MobileNetV2, OpenCV, SSD Face Detector

- Deployed a two-stage real-time pipeline: SSD-based face detection (Caffe model) followed by MobileNetV2 classification for masked/unmasked faces with confidence-labelled bounding boxes.
- Optimized for real-time throughput using batch inference across all detected faces per frame via webcam stream.

PCB Defect Detection System | YOLOv5s, 693 Images, 6 Defect Classes

- Built a real-time defect detection pipeline for smart manufacturing using YOLOv5s; designed 12-risk RMMM plan and sprint-based test case framework for production readiness.

AI Culture Storytelling Generator | Gemini 1.5 Flash, Streamlit, gTTS

- Built a multimodal Streamlit app that accepts image uploads and generates 200-300 word contextual stories using the Gemini 1.5 Flash vision-language model.
- Integrated Google Text-to-Speech (gTTS) for audio narration and provided downloadable story output.

RESEARCH IN PROGRESS

AI & GIS-Based Route Planning for Defense Navigation | DRDO CAIR

- Prototyping RL and graph-based path planning models under real-world spatial and environmental constraints; benchmarking efficiency against classical planning baselines.

Deep Learning for Skin Cancer Detection | TensorFlow, EfficientNetB3, Grad-CAM, Flask

- Developed a multi-class dermoscopic image classifier using EfficientNetB3 with transfer learning; achieved AUC 0.826 and sensitivity 0.916 across lesion categories, with focal loss to handle class imbalance.
- Diagnosed and resolved critical preprocessing mismatches (double rescaling, Keras 3 incompatibilities) and integrated Grad-CAM for spatial interpretability of model predictions.
- Deployed as a Flask web application with a heatmap grid visualization pipeline for per-class correctness analysis.

ACHIEVEMENTS & LEADERSHIP

Associate Lead - SRMKZILLA | *Open-Source & Tech Community*

2023 - Present

- Leading technical initiatives and open-source contributions at SRM IST's developer community.

TECHNICAL SKILLS

Core Areas: Computer Vision, Deep Learning, Metric Learning, Medical Imaging, NLP, AI Navigation, HRI

ML/DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers

CV & Vision Tools: OpenCV, MTCNN, Grad-CAM, DICOM, YOLOv5, EfficientNet, ResNet, MediaPipe

Programming: Python, C++, Java, SQL, JavaScript

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Dev & Deployment: Flask, Streamlit, Git/GitHub, FastAPI, Docker, Redis

Databases: PostgreSQL, MySQL, Firebase

Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, Statistics